SD FC Pricing Intelligence Platform

COO Interview Prep | VP of Data Analytics & Strategy | Q&A Reference

This document anticipates technical, business, and strategic questions a COO would ask about the pricing intelligence platform. Each answer is framed from the perspective of the VP candidate presenting this as their work product.

1. Model & Methodology

Q: Walk me through the model. How does it actually decide on a price?

It's a three-layer system. **Layer 1** is an XGBoost demand model that forecasts expected attendance (0–100%) for every section × game combination, using 60+ features across 14 groups — schedule, opponent strength, weather, secondary-market signals, inventory velocity, social sentiment, and cross-border demand. **Layer 2** is a causal elasticity model using Double Machine Learning (DML) that isolates the true price-to-demand relationship per seating tier, controlling for confounders like rivalry status and weather. This tells us *how much* demand drops per dollar of price increase. **Layer 3** is an EMSR-b revenue optimizer (the same algorithm airlines use for seat pricing) that finds the revenue-maximizing price for each section, then applies strategic guardrails — STH resale protection, fan-equity caps, backlash risk flags — to produce three scenarios: Conservative (0.5×), Balanced (1.0×, recommended), and Aggressive (1.5×).

Interview tip: Emphasize the guardrails layer — COOs care that the model won't embarrass the club.

Q: What are the most important features driving the model's predictions?

The top drivers (by SHAP feature importance) are: **(1) Secondary market premium** — the ratio of resale price to face price is the single strongest signal. If StubHub transactions are clearing at 40% above face, the market is telling us we're underpriced. **(2) Opponent tier and rivalry status** — Baja Cup matches against Club Tijuana drive 2–3× the demand of a midweek match against a bottom-table team. **(3) Sell-through velocity at T-7** — how fast seats are selling in the final week is a strong real-time demand signal. **(4) Day of week and season progress** — Saturday games significantly outperform Wednesday games; late-season games with playoff implications spike. **(5) Cross-border index** — unique to San Diego: ~6,000 regular fans cross from Tijuana, scaling to 2.5× for Baja Cup games. This is a feature no other MLS club needs.

Interview tip: Name the San Diego-specific features (cross-border, military market) — it shows you built this FOR this club, not as a generic tool.

Q: How do you know the model is accurate? What's your error rate?

We target a Mean Absolute Percentage Error (MAPE) under 15% on demand forecasts, validated using time-series cross-validation with a 14-day purge gap between folds — this prevents data leakage from sequential games. The MAPE tracks at around 8–12% in testing. But more importantly for business decisions, I built a **sensitivity analysis** that asks the real COO question: *what if the model underestimates how much demand drops when we raise prices?* Even if actual price elasticity is 50% steeper than our estimate, the Balanced scenario still delivers +2.6% uplift. The model's elasticity would need to be **2× wrong** before flat pricing wins. The model doesn't need to be perfect — it just needs to be directionally right.

Interview tip: Lead with the sensitivity analysis, not MAPE. COOs don't think in MAPE; they think in 'what if you're wrong?'

Q: The sensitivity table — how confident can we be in the uplift numbers?

The sensitivity table stress-tests the model's key assumption: **price elasticity**. Our model estimates how much demand drops per dollar of price increase (e.g., –0.58 for lower bowl midfield). The table asks: what if fans are actually *more* price-sensitive than we think? Results: at 50% elasticity error, Balanced still delivers +2.6% uplift. At 2× elasticity error (model is 100% wrong about price sensitivity), we break even with flat pricing. This is a **very wide margin of safety** — academic elasticity estimates for sports tickets typically have standard errors of 15–25%, not 100%. The Conservative scenario is even safer: it barely moves prices, so even extreme elasticity error doesn't hurt. The Aggressive scenario gives the most upside (+7.8%) but is the first to lose if elasticity is badly wrong. That's why we recommend Balanced as the default.

Interview tip: Frame it as: 'The model needs to be directionally right, not precisely right. Even if we're 50% off on elasticity, we still win.'

Q: Why did you choose XGBoost over deep learning or a simpler model?

Three reasons. **(1) Interpretability** — XGBoost produces SHAP explanations for every prediction, so when pricing ops asks 'why is the model recommending +$12 for this section?', we can show the exact feature contributions. A neural net is a black box. **(2) Sample size** — with ~238 section×game rows per season, we don't have millions of records. XGBoost handles small data well with proper regularization (max_depth=5, min_child_weight=3, gamma=0.1). A deep model would overfit. **(3) Production simplicity** — XGBoost runs in milliseconds, serializes to a small file, and doesn't require GPU infrastructure. For a first-year MLS club, operational simplicity matters.

Q: What is Double Machine Learning and why use it for elasticity?

Standard regression would give us a *correlation* between price and demand, which is biased: we raise prices for popular games (rivalry, marquee), so naive regression would show higher prices = higher demand. DML solves this by **partialing out confounders**. Step 1: predict price from confounders (opponent, weather, significance) — take the residual. Step 2: predict demand from the same confounders — take the residual. Step 3: regress demand residual on price residual — this gives the **causal** price effect. The elasticities range from –0.35 (field club, very inelastic — committed premium buyers) to –1.10 (upper concourse, very elastic — price-sensitive fans). These match academic sports economics literature and our tier priors, which gives us confidence the estimates are real.

Interview tip: If they push on 'is it really causal?', be honest: it's the best we can do with observational data. A true RCT would require randomizing ticket prices, which creates legal and PR risk. DML is the industry gold standard for this constraint.

2. Data & Accuracy

Q: What data sources does this use? Is any of it real-time?

Currently the platform ingests 11 data sources: Ticketmaster (face prices), StubHub/SeatGeek/VividSeats (secondary market transactions and listings), OddsPortal (betting odds as a proxy for expected match quality), FBRef (team performance metrics like xG), weather forecasts, and our internal inventory snapshots. Right now, the pipeline runs in **batch mode** — features are computed and the model re-scores weekly. It is not real-time. The architecture supports a shift to near-real-time (API endpoints already exist), but for Year 1, batch is the right trade-off: it keeps the system simple, lets pricing ops review recommendations before they go live, and avoids the operational risk of fully automated price changes during a club's first season.

Interview tip: Be upfront that it's batch, then frame it as a deliberate Year 1 choice. The roadmap to real-time is clear.

Q: You're using synthetic data. How can you trust the results?

Good question — this is the bootstrap problem every new franchise faces: you can't build a model without data, and you can't collect data without a season. Here's how we handled it: **(1)** The synthetic data is calibrated against real MLS benchmarks — league-wide attendance distributions, published secondary market premiums from StubHub, actual weather data for San Diego, and real 2025 betting odds from OddsPortal. **(2)** We anchored face prices to Ticketmaster's published tier structure. **(3)** The secondary market premiums follow established patterns from comparable expansion clubs (Charlotte FC, St. Louis CITY). **(4)** Most importantly, the model architecture is designed to **retrain on real data** as soon as the 2025 season provides actual transactions. The synthetic data is scaffolding — the value is in the pipeline, the feature engineering, and the optimizer logic, which all transfer directly to real data.

Interview tip: Acknowledge it openly, then redirect to the pipeline value. The synthetic data is ~80% of the questions about trustworthiness.

Q: How would you validate this with real data once the season starts?

Four-stage validation plan: **(1) Holdout backtesting** — after 5–6 home games (~70 section×game observations), retrain the demand model on games 1–4 and predict game 5–6. Compare predicted vs actual attendance by section. **(2) A/B price testing** — pick 2–3 comparable sections and apply dynamic pricing to one, keep the other at flat pricing. Measure revenue lift after 4–5 games. **(3) Secondary market tracking** — monitor whether the gap between face and resale narrows as we implement recommendations, which would confirm we're capturing the right price points. **(4) STH satisfaction survey** — quarterly check that season ticket holders can still resell at a profit, because if STH margins go negative, renewal rates collapse and the long-term economics fail.

Q: What happens if the model is wrong on a specific game?

The system has multiple safety nets. **First**, pricing ops always has final approval — no price goes live without a human reviewing the recommendation. **Second**, the guardrail layer caps maximum price increases at 25% before flagging for manual review (backlash risk threshold). **Third**, floor and ceiling prices per tier ensure we never go below cost or above market-breaking levels (e.g., supporters GA is capped at $18–$45). **Fourth**, the STH resale guarantee ensures primary price never exceeds 90% of the secondary sold average, so season ticket holders are always protected. If all of that fails on a specific game, the blast radius is one section on one night — it's not a systemic risk.

Interview tip: COOs think in risk management. List the guardrails as defense-in-depth layers.

3. Business Impact & Revenue

Q: Where does the $796K revenue opportunity come from?

It comes from the price gap analysis on the 2025 season. For every section × game, we compare the face price to the optimal face price implied by secondary market clearing prices. Formula: optimal_face = sold_price_avg / 1.15. The gap between that and the actual face price, multiplied by remaining seats, gives the revenue left on the table. Summing across all 238 section×game combinations for 2025: **$796K**. The bulk comes from high-demand games — Club Tijuana alone accounts for ~$109K because secondary prices cleared at 40–58% above face. The opportunity isn't everywhere equally — about 53% of section×game combinations are 'healthy' or 'hot' (demand at or above face), and that's where the money is.

Q: What does +5.3% revenue uplift actually mean in dollars?

For the 2025 season, total baseline revenue (face price × capacity × demand) across all section×game combinations is approximately $35.2M. A 5.3% uplift on that is roughly **$1.9M in additional revenue** per season. For context, that's the salary equivalent of 2–3 quality MLS players, or the operating budget for an entire analytics department for 5+ years. And this accounts for demand elasticity — some fans will buy less when prices go up, and the model factors that in. The Aggressive scenario projects +7.8%, though we recommend Balanced because it protects STH margins and minimizes backlash risk.

Q: How does this compare to what other MLS clubs are doing?

Most MLS clubs use rudimentary dynamic pricing — typically 3–4 manual price tiers set by a revenue manager based on perceived demand. A few clubs (Atlanta United, LAFC, Seattle) have invested in more sophisticated systems, but the industry is still early. What differentiates this platform: **(1)** Section-level granularity (most clubs price at tier level only, missing intra-tier variation). **(2)** Causal elasticity estimation (most clubs use rules of thumb, not econometric models). **(3)** STH health as a first-class constraint (most systems optimize revenue without protecting the resale market). **(4)** San Diego-specific features like cross-border demand and military market scoring that no off-the-shelf vendor provides.

Interview tip: Position as 'best-in-class for a Year 1 club' — don't oversell vs clubs with 10 years of real data.

Q: What's the cost to implement this? What's the ROI?

Infrastructure cost is minimal: the entire stack runs on open-source tools (Python, XGBoost, DuckDB, Streamlit). Cloud hosting is ~$50–200/month. The primary cost is people — one senior analytics hire (the VP role) plus one pricing analyst to review and action recommendations. All-in, call it $300–400K/year in fully loaded headcount + tools. Against a projected $1.9M revenue uplift in Year 1 (Balanced scenario, elasticity-adjusted), that's a **5:1 ROI in Year 1**, growing as the model improves with real data. By Year 2–3 with real data, the uplift should increase to 8–12% as the demand model tightens and we add real-time secondary market feeds.

4. Stakeholder Management & Risk

Q: How do you avoid negative PR from dynamic pricing? Fans hate being charged more.

This is the #1 risk and I built the system around it. Three strategies: **(1) Never call it 'dynamic pricing' externally.** The language matters — we frame it as 'market-based pricing' or 'demand-based seating tiers.' Airlines and hotels normalized this; sports clubs haven't yet, so the framing has to be careful. **(2) The backlash risk flag.** Any section×game where the recommended increase exceeds 25% gets flagged for manual review. We don't blindly push a 40% increase on a rivalry game — that's a social media crisis waiting to happen. **(3) Protect the STH first.** Season ticket holders are the economic backbone. The model guarantees they can resell at a net profit after marketplace fees. If a price increase would make an STH underwater on resale, the model won't recommend it. Happy STHs don't complain publicly and their renewal rate stays high.

Interview tip: This is likely the COO's top concern. Lead with empathy for the fan experience, then show the safeguards.

Q: How do you get buy-in from internal stakeholders — ticket ops, marketing, the head coach's staff?

The dashboard is designed as a **collaboration tool, not a dictation tool**. **Ticket Ops**: The Pricing Workshop view lets them test any price point with a slider and immediately see the impact on revenue, attendance, and STH margins. They keep final say. **Marketing**: The STH Value Dashboard shows which games have healthy resale margins (good for renewals) and which are cold (opportunities for targeted promos). They can use it for campaign planning. **Finance/FP&A**: The Performance Report gives them sensitivity analysis and revenue projections they can put into board decks. The key principle: **transparency builds trust**. Every recommendation comes with a SHAP-based plain-English explanation ('this price increase is driven by Club Tijuana rivalry + Saturday + 85% sell-through at T-7'). Nobody has to take the model's word for it.

Q: What if season ticket holders complain that dynamic pricing hurts their resale value?

The system is explicitly designed to **protect STH resale economics**. The STH resale guarantee guardrail ensures that primary prices never exceed 90% of the average secondary market clearing price (after accounting for ~10% marketplace fees). This means an STH can always resell at a net profit on hot games. The dashboard tracks this in real time — the STH Value Dashboard shows net margin per seat, per game, after fees. For 2025, the average STH net resale value is $204/seat across the season, with only 4 of 17 games showing negative margins (cold midweek games — exactly the ones STHs would attend themselves rather than resell). If anything, smart dynamic pricing **helps** STHs: by pricing hot games closer to market clearing, we reduce the gap that scalpers exploit, which means more of the value stays in the STH ecosystem.

Interview tip: 'We're protecting STHs FROM the scalpers, not competing with them' is a powerful reframe.

Q: What's the worst-case scenario if this goes wrong?

The worst case is a visible pricing error on a high-profile game — say, dropping supporters section prices to $18 for a rivalry game (model underpredicts demand) or pushing premium seats to $400 on a Wednesday game nobody wants. Mitigation: **(1)** Human review before any price goes live. **(2)** Hard floor and ceiling prices per tier ($18–$45 for supporters, $140–$400 for field club). **(3)** The 25% backlash threshold. **(4)** We start with the Balanced scenario, not Aggressive, keeping adjustments moderate. **(5)** We can roll back to flat pricing on any game within minutes — the system is advisory, not automated. Financial worst case: if the model is consistently wrong (MAPE >25%), we're back to flat pricing with no uplift — but we haven't lost money, just the opportunity cost. The downside is bounded; the upside is not.

Q: Are there legal or compliance considerations with dynamic pricing?

Yes, a few. **(1) State consumer protection laws** — California doesn't prohibit dynamic pricing for events (it's legal), but we need clear disclosure at point of sale that prices are demand-based. **(2) ADA accessibility** — accessible seating must be available at the lowest price offered for the section; we enforce this in the guardrail layer. **(3) Sponsor/partner contractual pricing** — some sections may have fixed-price allocations for sponsors; the model excludes those from optimization. **(4) MLS league policies** — the league encourages revenue optimization but has soft guidance against price gouging on rivalry games. Our backlash risk flag handles this.

5. Real-World Testing & Implementation

Q: How would you test this in real life before fully rolling it out?

Phase 1 (Games 1–4): Shadow mode. Run the model on every game but don't change any prices. Compare what the model would have recommended vs what actually happened. Track prediction accuracy and build confidence. **Phase 2 (Games 5–8): A/B test on 2–3 sections.** Pick comparable sections (e.g., two upper bowl sections with similar demand profiles). Apply dynamic pricing to one, keep the other flat. Measure revenue, attendance, and STH feedback. **Phase 3 (Games 9+): Gradual rollout.** If the A/B shows positive lift with no backlash, expand to all sections using the Conservative scenario first, then graduate to Balanced. **Phase 4 (Season 2): Full deployment.** By now we have 17 games of real data to retrain on. Move to Balanced scenario across the board with weekly model refreshes.

Interview tip: A phased rollout shows operational maturity. COOs love 'crawl, walk, run' language.

Q: What KPIs would you track to know if this is working?

Five primary metrics: **(1) Revenue per game vs flat-price baseline** — the headline number. Target: +5–10% in Year 1. **(2) Demand forecast MAPE** — model accuracy. Target: <15%, ideally <10%. **(3) STH net resale margin per game** — must stay positive on 80%+ of games. If this drops, we're over-pricing and hurting the renewal pipeline. **(4) Fan sentiment score** — social listening + CSAT surveys. If dynamic pricing discussions go negative, we dial back. **(5) Attendance rate vs optimal capacity** — are we filling seats or just raising price on empty sections? Revenue optimization with 70% attendance is worse than moderate pricing at 95%. Secondary metrics: secondary market gap closure, price change acceptance rate by section, and guardrail trigger frequency (if >30% of recommendations hit guardrails, the model needs recalibration).

Q: Is this system real-time, or does someone have to run it manually?

Currently it runs in **batch mode** — the full pipeline (feature engineering → model scoring → price recommendations) executes on-demand or on a weekly schedule. The results feed into the Streamlit dashboard where pricing ops reviews and approves changes. This is a **deliberate Year 1 choice**. Real-time pricing requires: (a) live inventory APIs, (b) automated price push to Ticketmaster, (c) incident response for bad recommendations. For a club's first season, adding that operational complexity creates more risk than value. The **roadmap to real-time** is: Year 2 — near-real-time secondary market feeds with 4-hour refresh; Year 3 — automated Ticketmaster API integration with human-in-the-loop approval workflow; Year 4+ — fully automated with exception-based human review only.

6. Strategic Vision & Growth

Q: Where does this go after Year 1? What's the bigger vision?

This ticket pricing module is **Phase 1 of a broader sports business intelligence platform**. The roadmap: **Phase 2 — Sponsorship valuation:** Use the same demand and attendance models to price sponsorship assets (LED boards, jersey patches, in-venue activations) based on projected eyeballs per game instead of flat annual rates. **Phase 3 — Concessions & merchandise optimization:** Dynamic pricing and inventory optimization for F&B based on expected attendance and dwell time. **Phase 4 — Fan lifetime value model:** Predict which single-game buyers will convert to STH, what price sensitivity they have, and optimize the conversion funnel. **Phase 5 — Multi-club platform:** Package the pipeline as a product for other MLS/USL clubs. The feature engineering and optimizer are parameterized — swap in a new club's data and the system works out of the box.

Interview tip: Show you're thinking beyond one module. VP of Strategy should think in platforms, not projects.

Q: How would you upgrade the model to be more accurate over time?

Four concrete upgrades as real data flows in: **(1) Real transaction data retraining** — after each game, actual attendance and secondary market transactions feed back into the model. XGBoost retrains weekly with expanding window. MAPE should drop from ~10–12% to 6–8% by mid-season. **(2) Live secondary market ingestion** — replace batch StubHub snapshots with streaming SeatGeek/VividSeats APIs. Price signals at T-48hrs and T-24hrs are the most predictive. **(3) Weather API integration** — currently using historical San Diego weather patterns. Switching to 10-day NWS forecasts for game-specific rain probability would improve the weather demand impact feature. **(4) Social sentiment from real sources** — currently estimated. Integrating Reddit, Twitter/X, and team Discord activity as a feature (social NLP scores) gives a 2–3% MAPE improvement based on academic benchmarks.

Q: Can this work for other sports or other clubs?

Yes. The architecture is intentionally modular: **What transfers directly:** the three-layer pipeline (demand model → elasticity → optimizer), the guardrail framework, and the dashboard. These are sport-agnostic. **What needs recalibration per club:** feature engineering (different sports have different demand drivers — NFL weather matters more, NBA back-to-backs matter, etc.), elasticity priors (premium NBA sections may have different elasticity than MLS), and guardrail thresholds. **What's San Diego-specific:** the cross-border features (Tijuana market), military market scoring, and Baja Cup variables. These would be replaced with local market features for another city. The total integration time for a new MLS club would be ~4–6 weeks if they have Ticketmaster and secondary market data accessible.

Q: What team would you need to build around this function?

Lean and scalable: **Year 1 (2 people):** VP of Strategy & Analytics (this role — owns the model, strategy, and executive communication) + one Pricing Analyst (daily ops, reviews recommendations, coordinates with ticket ops, monitors model health). **Year 2 (3–4 people):** Add a Data Engineer (build real-time pipelines, maintain infrastructure) and a Business Intelligence Analyst (expand dashboard to sponsorship and concessions). **Year 3+ (5–6 people):** Add a second ML Engineer for real-time model serving and a Fan Insights Analyst for LTV modeling and conversion optimization. The philosophy: start small, prove ROI, then invest. Don't build a 10-person team before you've proven the $1.9M uplift is real.

Interview tip: COOs appreciate hiring discipline. Show you won't ask for a big team on day one.

7. Your Background & Fit

Q: Why are you the right person for this role?

Three things make me uniquely suited: **(1) I've been a club CEO.** I don't just build models — I understand the P&L, the stakeholder dynamics, the fan politics, and the board-level conversations that determine whether a pricing initiative lives or dies. Most data scientists can build the model; very few can walk into a board meeting and explain why it won't cause a PR crisis. **(2) Technical depth with CS + MBA.** I can go from writing the XGBoost feature pipeline to presenting the ROI case to investors in the same meeting. This role sits at the intersection of strategy and analytics — you need someone who's native in both languages. **(3) This platform is my proof of work.** I didn't build a slide deck about what I *would* do. I built the entire system — data pipeline, ML models, causal elasticity estimation, revenue optimizer, interactive dashboard, and the business case — end to end. For San Diego FC specifically, not as a generic exercise. That's the level of ownership I bring.

Interview tip: Be specific about why club CEO experience matters in a VP analytics role — it's your biggest differentiator.

Q: What's the first thing you'd do in the first 90 days?

Days 1–30: Listen and validate. Meet with ticket ops, marketing, finance, and the coaching staff to understand their current process, pain points, and political dynamics. Run the model in shadow mode on the first 2–3 home games. Build relationships before pushing changes. **Days 31–60: Quick wins.** Present the shadow-mode results — 'here's what the model would have recommended, and here's the revenue we left on the table.' Get buy-in for an A/B test on 2–3 sections. Stand up the dashboard for ticket ops to play with. **Days 61–90: First results.** Run the A/B test, measure results, present to leadership. If positive, propose the Phase 3 rollout plan. Start scoping Phase 2 (sponsorship valuation) to show the bigger platform vision.

Q: How do you handle disagreements with ticket ops or other departments?

With data, transparency, and respect for their expertise. If ticket ops says 'the model is wrong on this game — trust me, this rival fills the house,' my response is: '*Show me what I'm missing.*' Maybe there's a local factor the model doesn't capture yet. That becomes a new feature. The model is a **tool for better decisions, not a replacement for human judgment**. If a 20-year ticket ops veteran has a strong conviction that contradicts the model on a specific game, we go with their call and track the outcome. Over time, the model earns trust by being right more often than not — but it never overrides institutional knowledge without a conversation. The worst thing an analytics leader can do is come in and say 'the model says you're wrong.' That's how you get fired and how analytics departments get disbanded.

Interview tip: Show emotional intelligence, not just analytical capability. COOs manage people, and they want a VP who can too.